

Lecture One

CSCI Technology Innovation | W26

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Thinking like an innovator

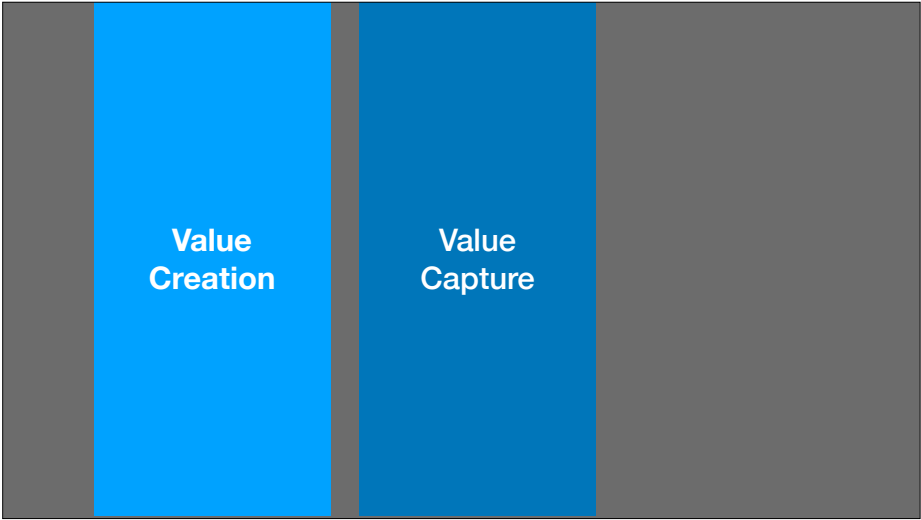
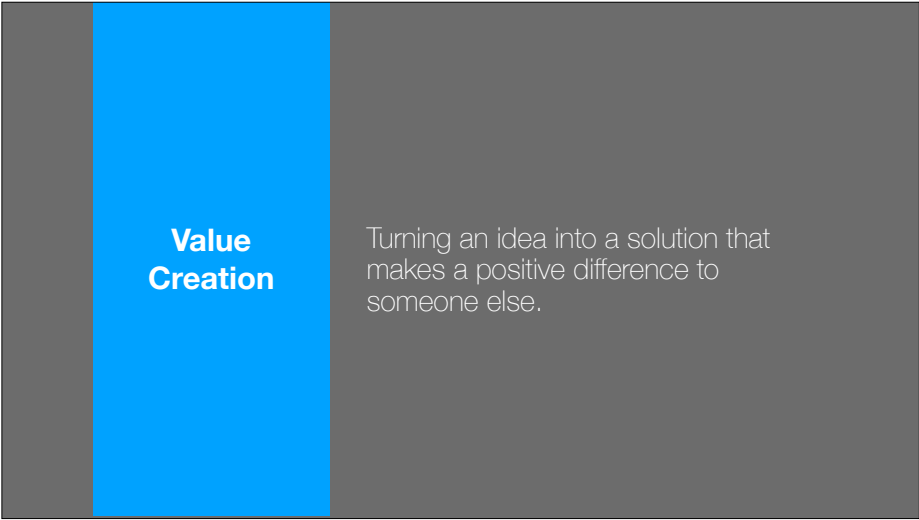
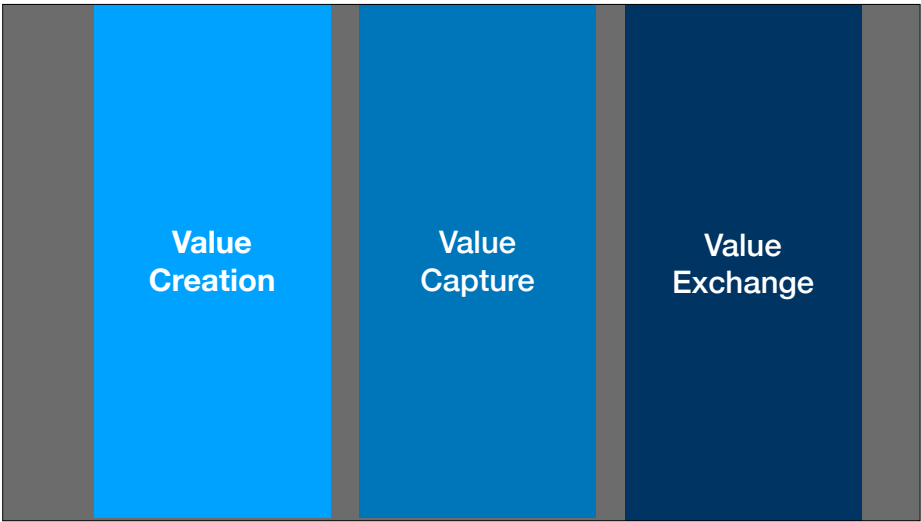
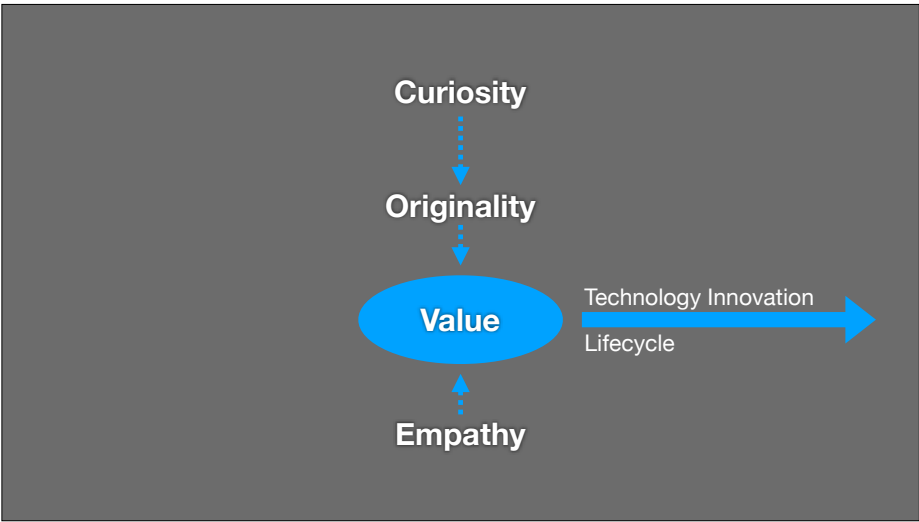
COVE

Curiosity

Originality

Value

Empathy



**Value
Creation**

**Value
Capture**



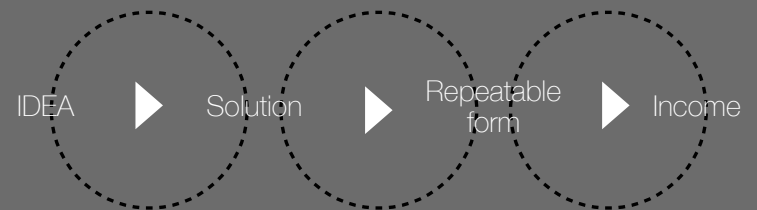
**Value
Creation**

**Value
Capture**

Giving the idea a form that others can see, use, and that you can protect (e.g., a product or a service).

Trading use of the idea into income.

**Value
Exchange**



**Value
Creation**

**Value
Capture**

**Value
Exchange**

“...the best (technology) leaders have the deepest understanding of the engineering and product design.”

- Larry Page

“... technology alone is not enough ... it's the technology married with liberal arts, married with the humanities, that yields the result that makes our hearts sing.”

- Steve Jobs

“poetic science”

- Ada Lovelace



The power of collaboration

Combining and sharing different skills and perspectives

Having one idea spark other ideas, over and over again

Value Creation

Where do great product ideas come from?

Value Capture

How can I make my idea real enough to show others?

How can I get make sure someone else won't take my idea for themselves?

Value Exchange

How can I make money from my idea?

Most successful technology innovators approach their work as a “product person.”

- Can I turn someone's problem in a product?
- Who has this problem?
- How can I know more about them?
- Can my idea do something for them that no other product can?
- What deeper need does it solve?

Course Concept #1

Creative Confidence

Being comfortable showing new, different, or incomplete work.

Course Concept #2

Build - Measure - Learn

Continuous trial and error.

Course Concept #3

**Falling in love with the
problem, not the
solution**

Staying curious.

THE BIG IDEA

Technology innovation turns people's problems into valuable solutions. We do this with a process for curiosity, originality, and customer empathy, in continuous cycles.

Curiosity

EXAMPLE BRAINSTORMING METHODS

FIVE WHYS

CRAZY 8's

SCAMPER

NO NO's

Five Whys

FROM TOYOTA

1 - WHY DID THE ROBOT STOP?

The circuit was overloaded, causing a fuse to blow.

2 - WHY WAS THE CIRCUIT OVERLOADED?

There was insufficient lubrication on the bearings, so they locked up.

3 - WHY WAS THERE INSUFFICIENT LUBRICATION ON THE BEARING?

The oil pump was not circulating sufficient oil.

4 - WHY WAS THE PUMP NOT CIRCULATING SUFFICIENT OIL?

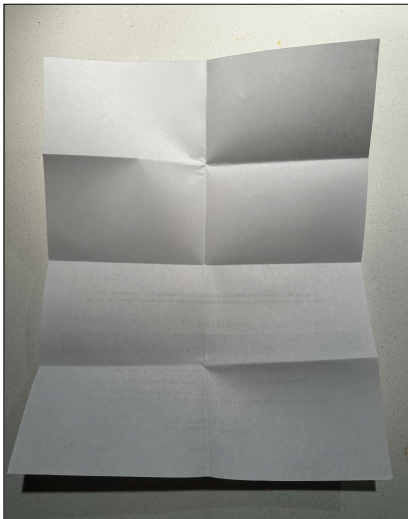
The pump intake was clogged with metal shavings.

5 - WHY IS THE INTAKE CLOGGED WITH METAL SHAVINGS?

There isn't a filter on the pump.

← Underlying problem to solve

CRAZY 8's



Eight ideas in eight minutes
High volume in short time